MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 2G21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode-heptode converter, subminiature

Outline --- 2-5 (EIA)

Base --- 7 pins in line

Envelope --- T2X3

Cathode --- Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	1a (triode) (red dot)	2a (heptode)	g1 (triode & heptode)	-f	2g3	2g2 2g4	+f 2g5

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	E1b	Ec1	E2b	E2c2+4	E2c3	Eg1	Ik	Rg1	Alt
Unit:	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Vac	mAdc	Meg	ft
Maximum:	1.50	50	---	50	50	Note 2	---	2.2	---	Note 3
	Note 1									
Minimum:	1.00	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	1.25	22.5	0	22.5	22.5	0	4.5	---	0.05	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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	METHOD	REQUIREMENT OR TEST	NOTES	CONDITION	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
		Quality conformance inspection, part 1								
(B)	1266	Total grid current	4	E2b = E1b = E2c2+4 = 30 Vdc; E2c3 = -1 Vdc	0.65	II	I2c3	0	-0.5	μAdc
	1256	Electrode current (heptode anode)	-		0.65	II	I2b	0.10	0.30	mAdc
	1256	Electrode current (screen)	-		0.65	II	I2c2+4	0.25	0.55	mAdc
	1256	Electrode current (oscillator anode)	-		0.65	II	I1b	0.5	1.0	mAdc
(B)	1237	Oscillator grid current	4	Zgk = 9,000 ohms; Ef = 1.0 Vdc	0.65	II	Ic1	20	---	μAdc
	1326	Conversion transconductance	-	Eg1 = 4.5 Vac; E2c3 = 0.0354 Vac	0.65	II	Sc	50	100	μmhos
	1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
		Quality conformance inspection, part 2								
	1301	Filament current	-		---	---	If	47	59	mAdc
	1031	Low-frequency vibration	-	R2p = 10,000 ohms; E2c3 = 0	---	---	Ep	---	200	mVac
	1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
	1119	Glass-to-lead seal strain	-		---	---	---	---	---	---
	1116	Lead fatigue	-		4.0	S3	---	---	---	---
	2126	Glass strain	-		4.0	I	---	---	---	---
	1331	Direct-interelectrode capacitance	5		---	---	C2g3-2a	---	0.10	pF
							C2g3-all	2.0	5.0	pF
							C2a-all	2.1	5.1	pF
	1105	Permanence of marking	-		---	---	---	---	---	---
		Quality conformance inspection, part 3								
	1501	Intermittent life	-	Group A; Ef = 1.25 Vdc or Vac with equivalent bias	---	---	---	---	---	---
	---	Intermittent life-test end points (500 hours):								
	1326	Conversion transconductance	-		---	---	Sc	30	---	μmhos
	1237	Oscillator grid current	-	Ef = 1.25 Vdc	---	---	Ic1	20	---	μAdc
	1266	Total grid current	-		---	---	I2c3	---	-1.0	μAdc
(B)	1202	Filament burnout	6		---	---	---	---	---	---

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NOTES:

1. Do not use in series filament circuits.
2. Never positive.
3. See "Reduced pressure (altitude) rating" paragraph in the basic document.
4. This test to be performed at the conclusion of the holding period.
5. With shield, per figure 1, connected to negative filament.
6. (B) This test shall be performed during the initial production and once each succeeding 12 calendar months in which there is production. A regular double sampling plan shall be used, with the first sample three (3) tubes with an acceptance number of zero (0), and a second sample of three (3) tubes with a combined acceptance number of two (2). In the event of failure, the test will be made as a part of QCI Part 2 Code Level D with a 6.5 AQL. The regular "12 calendar month" double sampling plan shall be reinstated after three (3) consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

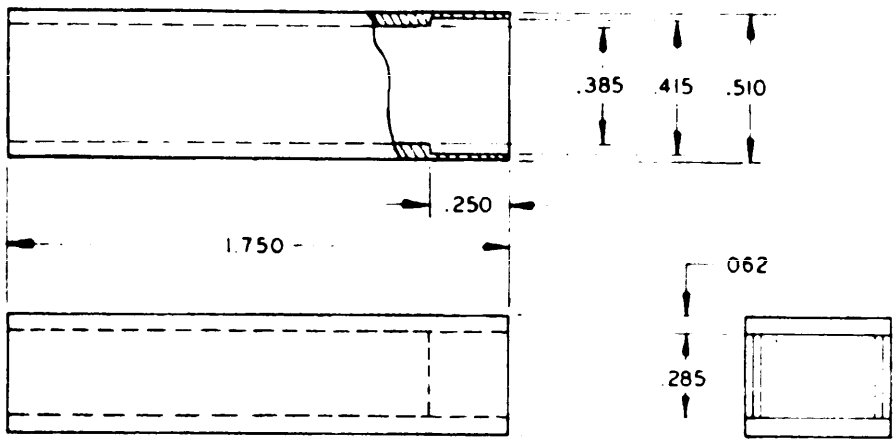
User activities:
Army - ME
Navy - AS, OS, MC, CG, SH

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5960-2910)

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NOTES:
a. All dimensions in inches.
b. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
c. Unless otherwise specified, tolerance is .xxx±.003, -.000.
d. Material: Brass, joints soldered.

INCHES	MM
.003	.08
.062	1.57
.250	6.35
.285	7.24
.385	9.78
.415	10.54
.510	12.95
1.750	44.45

FIGURE 1. T2 x 3 subminiature capacitance shield.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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Commanding General
U.S.Army Electronics Command
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/341B ELECTRON TUBE, RECEIVING TYPE 2G21

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
( ) DIRECT GOVERNMENT CONTRACT ( ) SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ( ) YES ( ) NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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